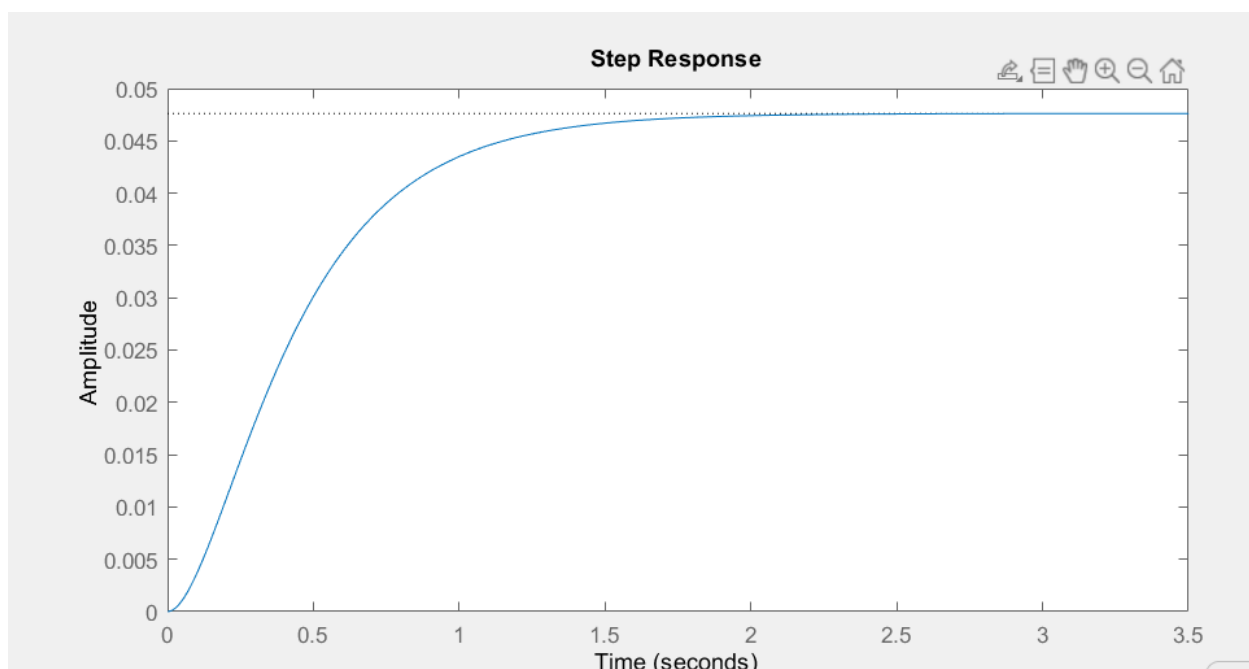


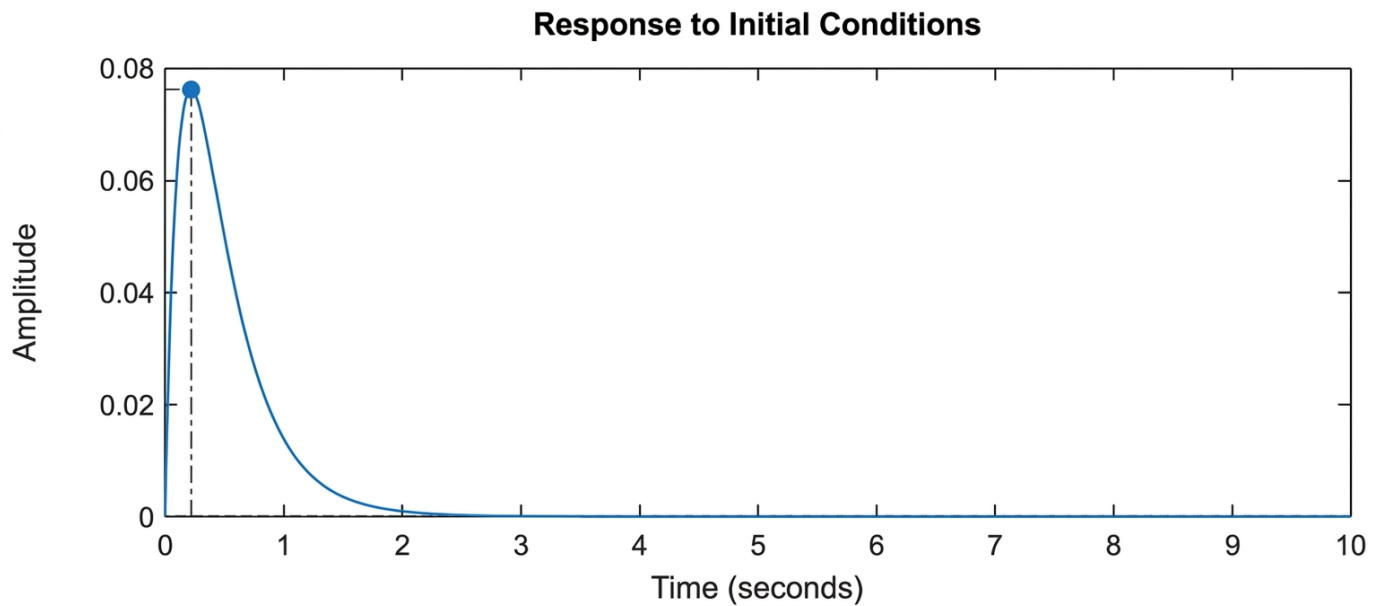
LAB – 10

In this lab, we first convert a transfer function into a state-space model in MATLAB by using the `tf2ss()` function. After that, we apply our work in Simulink.

```
vanid_Lab_10.m
1 -   clc ; clear all ; close all;
2 -   s=tf('s');
3 -   % from link
4 -   P = 1/(s^2 + 10*s + 20);
5 -   %den of our example
6 -   den = [1 10 20];
7 -   %state model from transfer function
8 -   [A,B,C,D]=tf2ss(1,den);
9 -   %convert it to system and apply step response.
10 -  sys = ss(A,B,C,D);
11 -  step(feedback(sys,1))
12 -  eig(sys)
13 -  eig(P)
```

This is the MATLAB code for the task. We define a transfer function called P. By using the `tf2ss()` function, we convert this transfer function into the matrices A, B, C, and D, which are used in the state-space model. To find the unforced (natural) response of the system, we can use the `initial()` function. For this, we choose an initial condition x_0 , as shown in the code above. The resulting graphs are shown below:

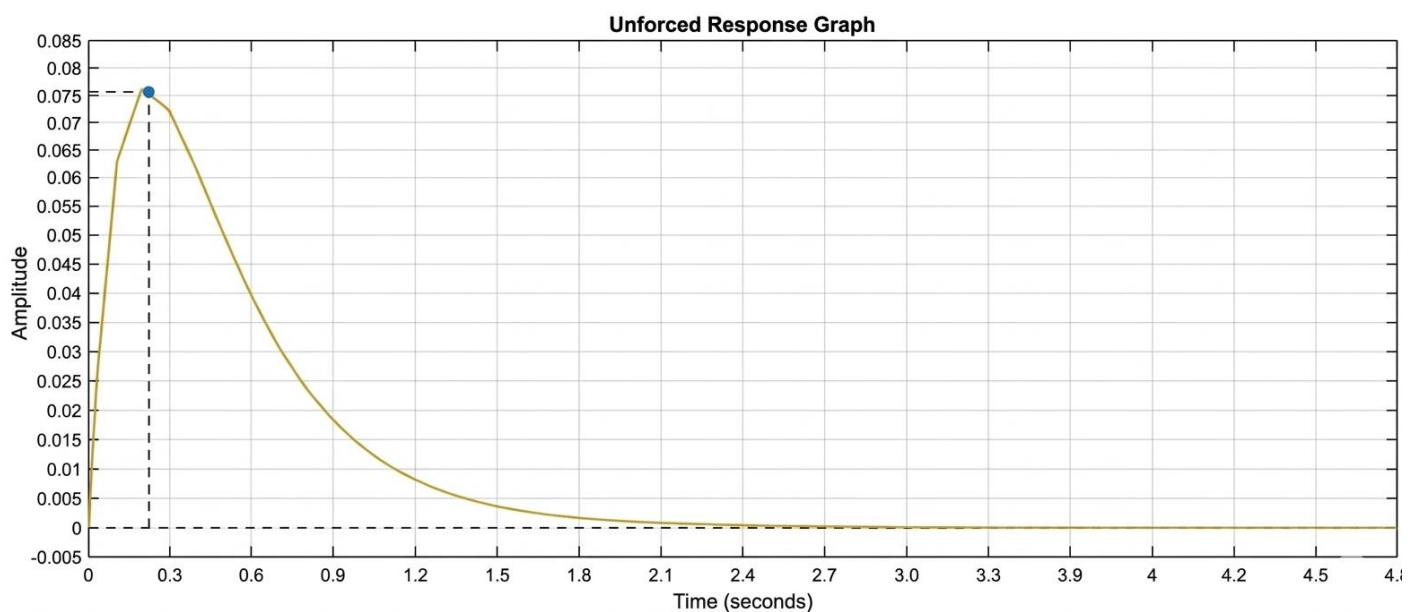




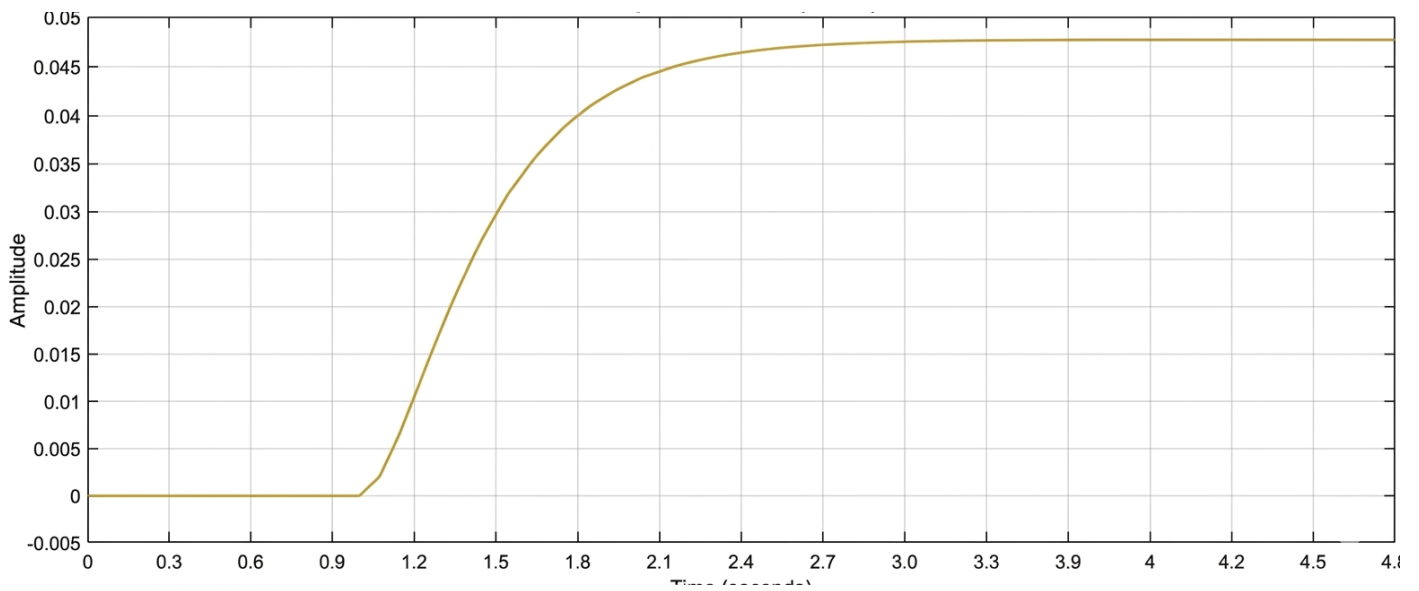
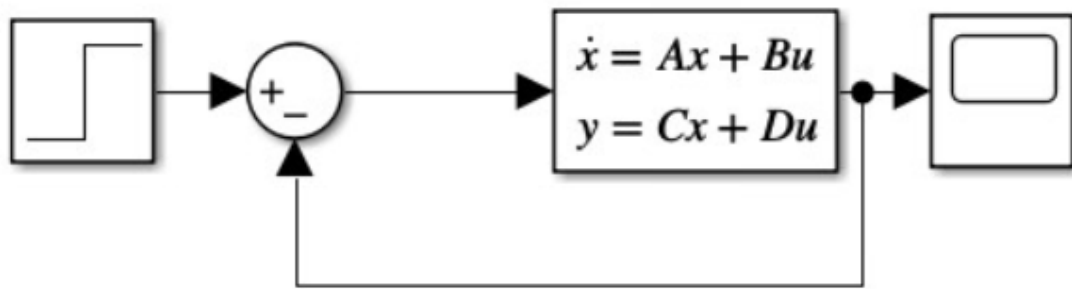
Now, for the implementation of MATLAB into Simulink, let's begin with unforced response:



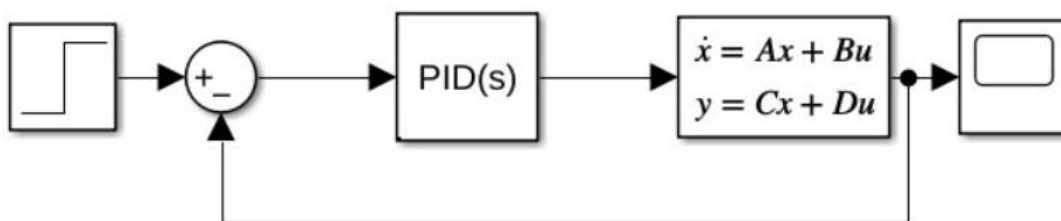
We make external input 0 and initial condition x_0 from the code. This will give us unforced response:



We have the condition with external input:



Last but not least, PID controller must be added to the system



Graph of the system with PID controller:

